Quantum Rings

Quantum Rings Workshop

iQuHack 2026

Bob Wold

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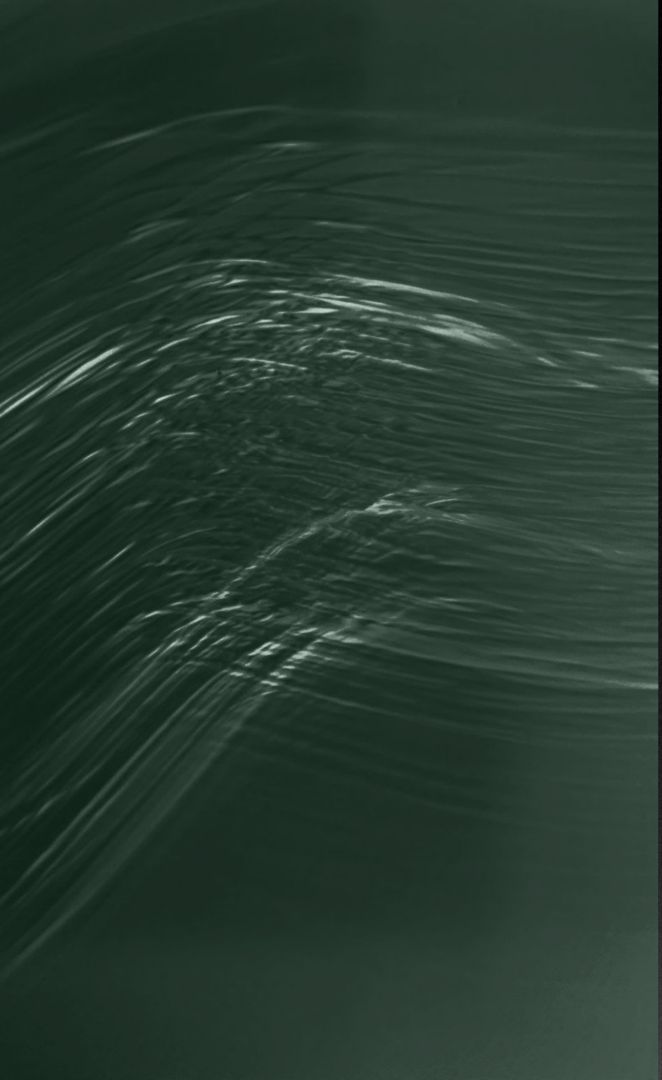
Rob Wamsley, PhD

Applications Lead
rob@quantumrings.com



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Market Context

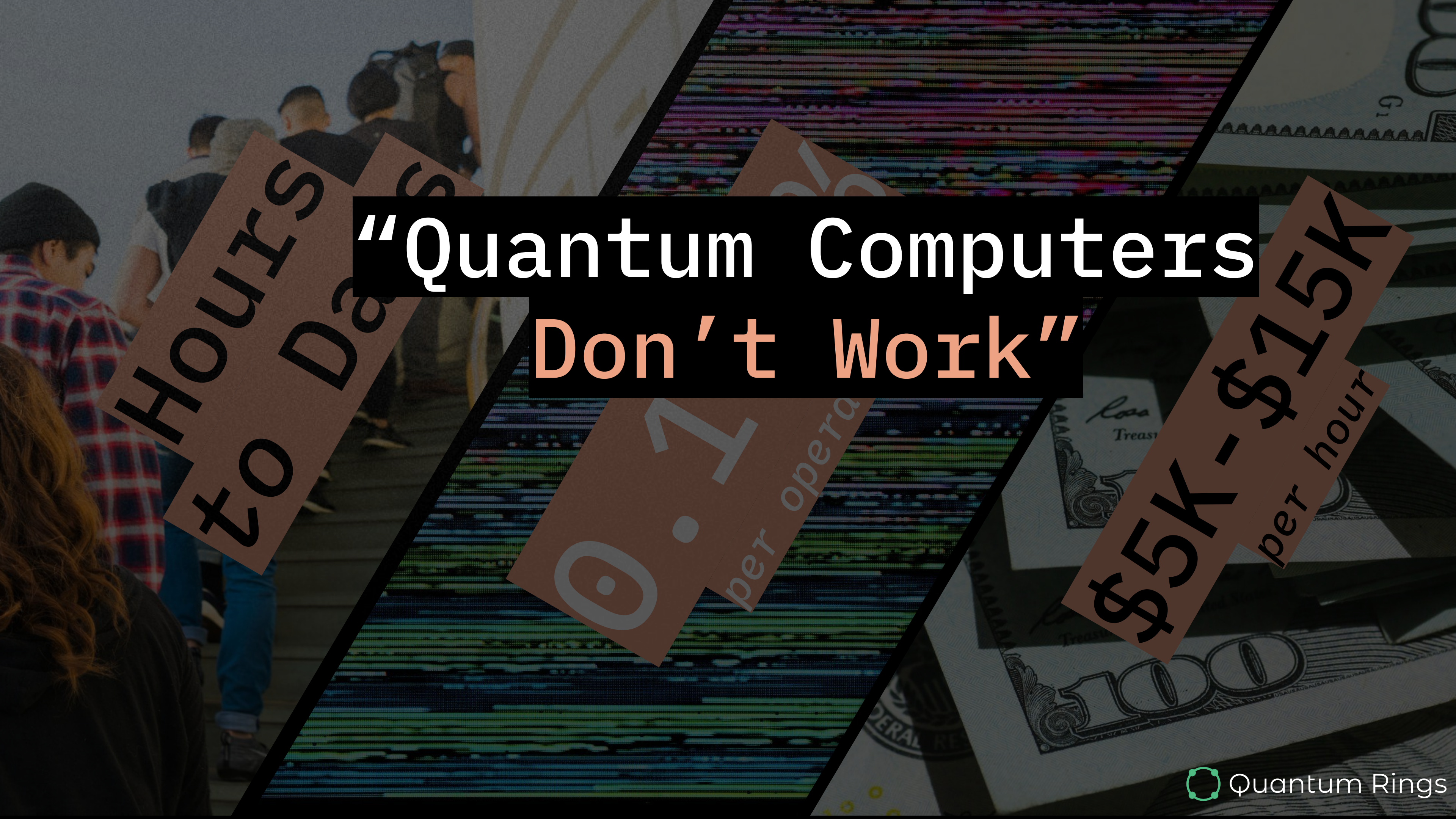
Quantum Rings

Quantum Computing Holds the Promise to Transform the World

Unlocking $2 TRILLION in Economic Value by 2035[1]

[1] Quantum Technology Monitor, McKinsey & Company

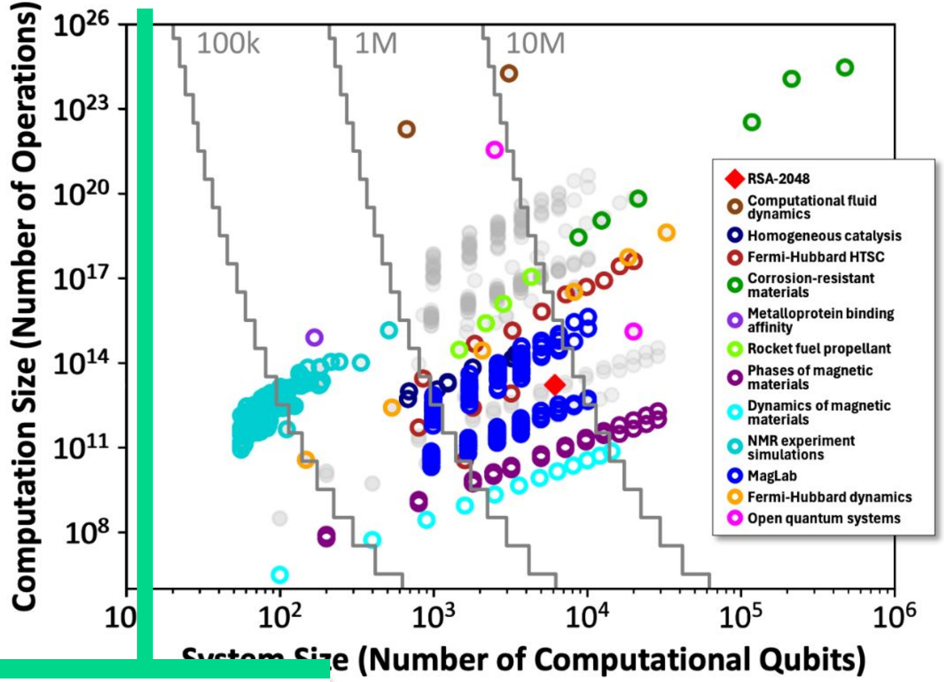
Quantum Rings


“Quantum Computers Don’t Work”
Hours
0.1
$5K-$15K
per hour
Quantum Rings

Preliminary results from Quantum Benchmarking

Applications that could benefit from a quantum coprocessor:

- Simulating correlated materials
- Developing corrosion resistant materials
- Developing new rocket fuels and explosive materials
- Dynamical simulations (for new solar cells, better understanding of biological processes, and magnetic materials)
- New methods to compile algorithms to fault-tolerant quantum architectures



...mistic resource estimates. Colored circles are optimistic resource estimates
All points supported by detailed published pre-prints.

Sept-2024


QUANTINUUM
HELIOS
The World's Most Powerful
Quantum Supercomputer
The Industry's First
50 Logical Qubit
Commercial System
Our IYQ Present To You!!
QUANTINUUM
Quantum world congress
Quantum Rings

Preliminary results from Quantum Benchmarking

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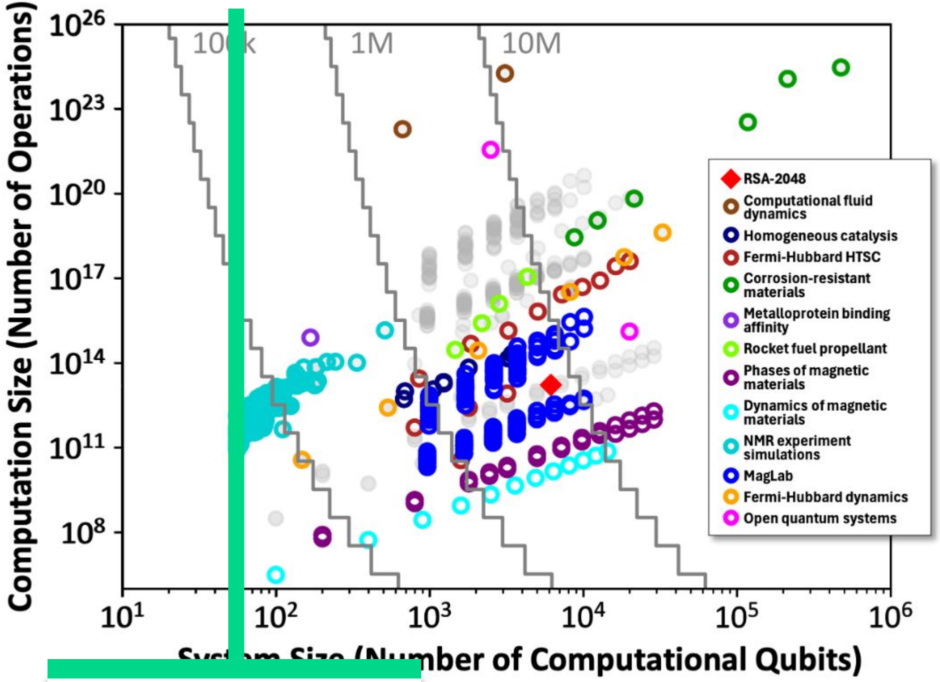
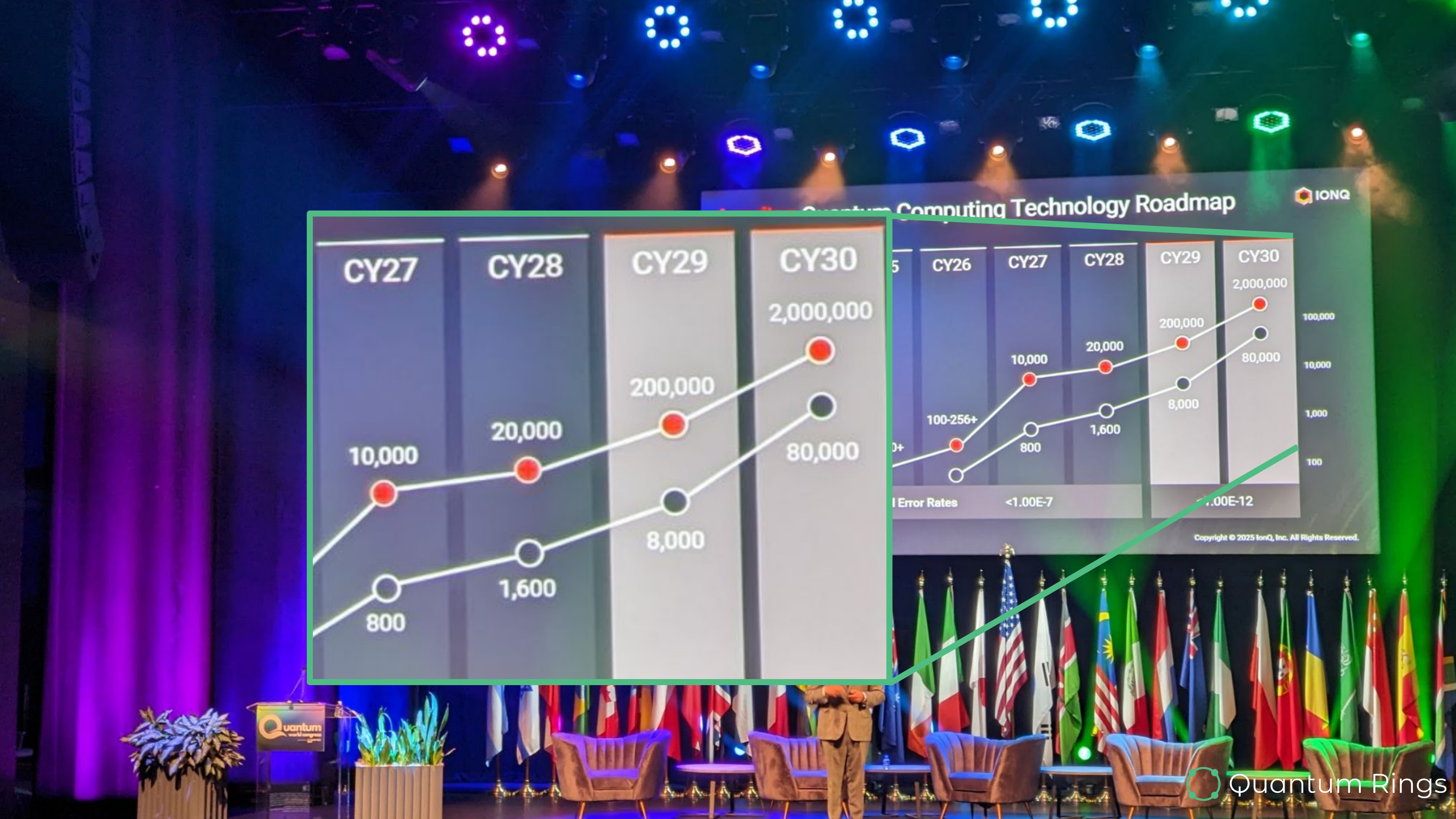


You Are Here

ce estimates. Colored circles are optimistic resource estimates
pported by detailed published pre-prints.

https://www.darpa.mil/sites/default/files/attachmen … rogram-manager.pdf

Quantum Rings


Computing Technology Roadmap
IONQ
CY27
CY28
CY29
CY30
2,000,000
200,000
20,000
10,000
80,000
8,000
1,600
800
CY26
CY27
CY28
CY29
CY30
2,000,000
200,000
20,000
10,000
100-256+
80,000
8,000
1,600
800
100,000
10,000
1,000
100
Error Rates
<1.00E-7
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Quantum Rings

Preliminary results from Quantum Benchmarking

Applications that could benefit from a quantum coprocessor:

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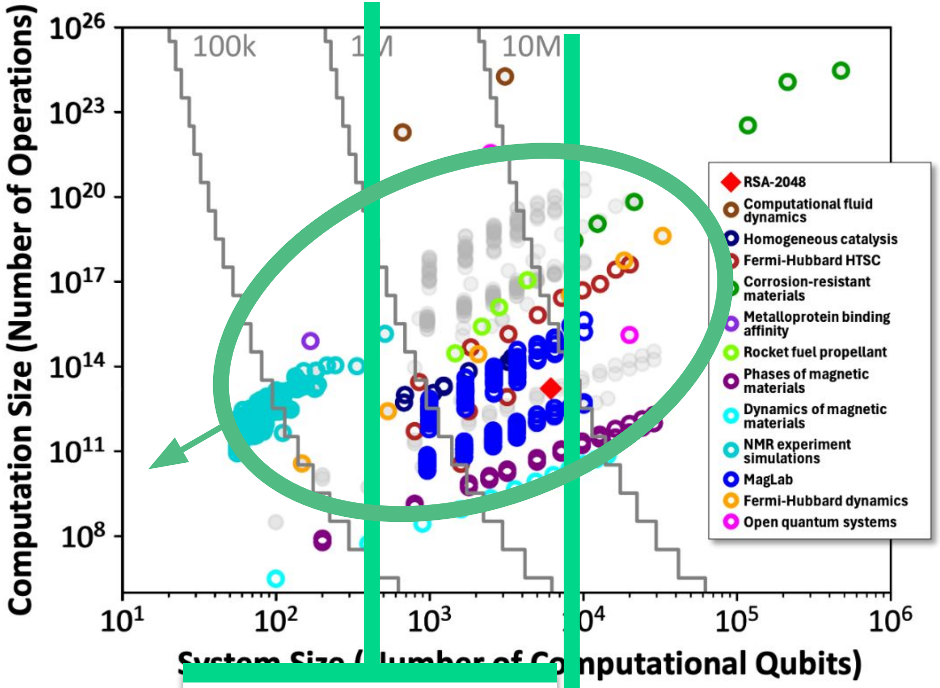


202

2029

Grey solid circl… resource estimates
based on know…

https://www.darpa.mil/sites/default/files/attachment/2024-12/qb

uantum Rings

August 13, 2024

NIST Releases Post-Quantum Encryption Standards

How Are Companies Responding?

Quantum Rings

Many Companies are Investing in Quantum Computing Programs Today

To name a few:

Chemicals

BASF

Dow

ExxonMobil

Life Sciences

Pfizer

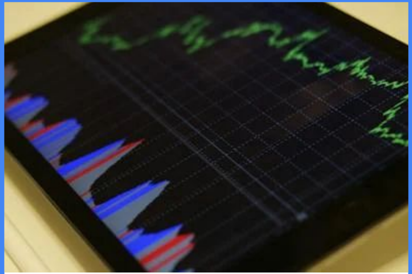
St. Jude Children's Research Hospital

Finding cures. Saving children.

ALSAC · DANNY THOMAS, FOUNDER

MERCK

Johnson & Johnson



Financial Services

JPMorgan Chase & Co.

Wells Fargo

StateFarm

citi

Transportation & Logistics + Auto

AIRBUS

BMW

FedEx

…More than many think.

60%

Fortune 500 Companies Actively Exploring Or Engaging With Quantum Computing [1]

98%

Will Maintain Or Increase Their Spend In Quantum Computing Next Year [3]

82%

Immaturity Of Quantum Computing Blocking Their Development [2]

[1] IBM Unveils $150 Billion Investment, IBM 2025
[2] Quantum Computing in Financial Services: A Business Leader's Guide, Moody's Analytic
[3] Quantum Computing Is Becoming Business Ready, BCG 2023

So… What are they doing?

Assessment Phase

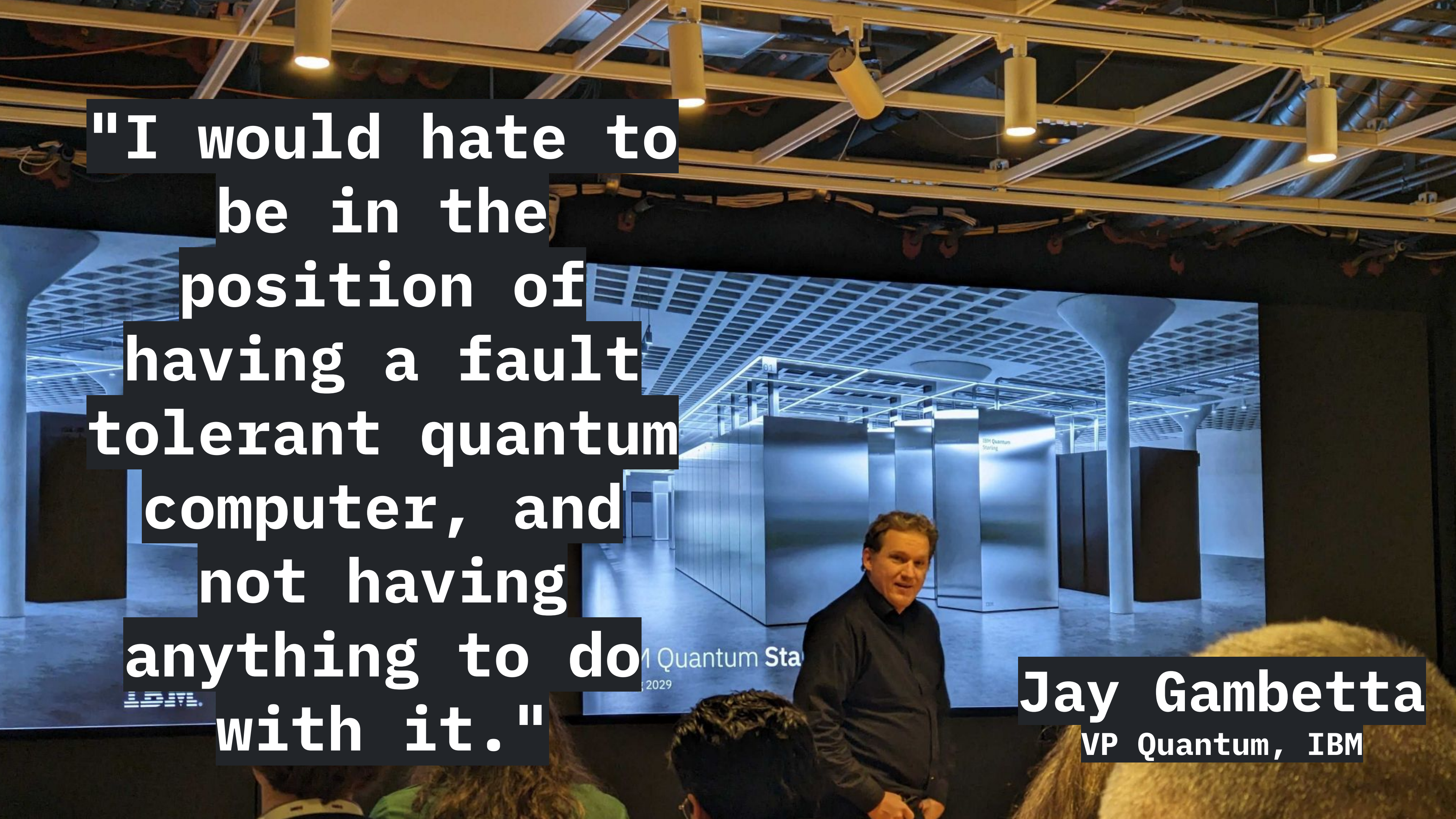
Algorithm + Problem = Application

QPU Experimentation

1. Strategy
Resource Estimates.
Time Horizons.

2. Innovation
Experimentation.

3. IP Harvesting



"I would hate to be in the position of having a fault tolerant quantum computer, and not having anything to do with it."

Jay Gambetta
VP Quantum, IBM

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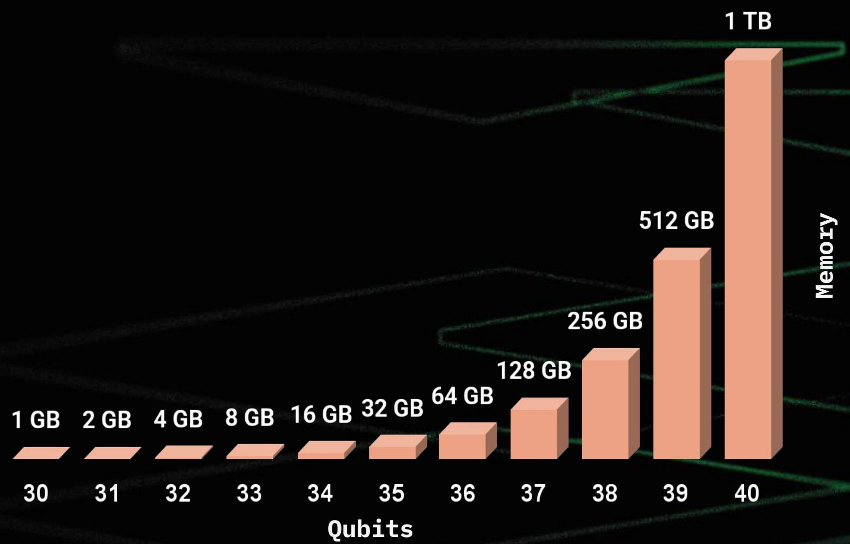
Quantum Rings

Quantum Rings

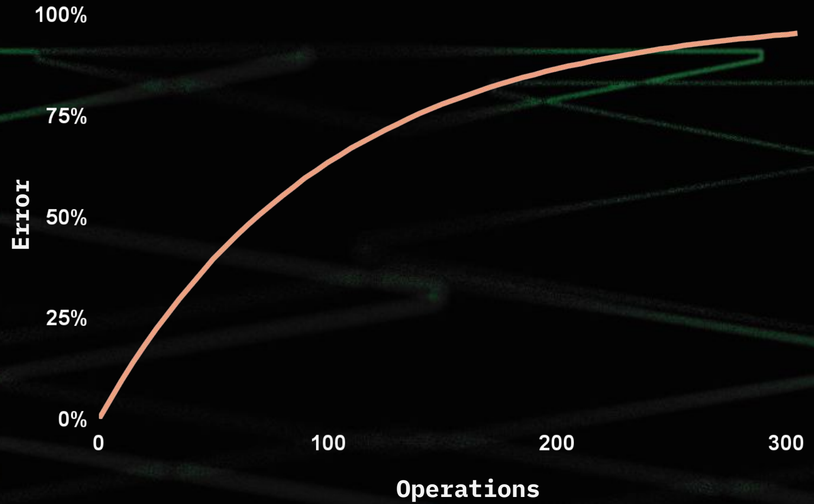
On a Mission to Democratize Quantum Development

Perfect Simulations



State Vector simulation is memory limited to ~35 qubits on a desktop. ~40 qubits on a supercomputer node.

Quantum Hardware



0.1% - 1% errors per operation prevent useful quantum circuits, while creating massive wait times (hours-days), and massive costs ($1.60/sec+)

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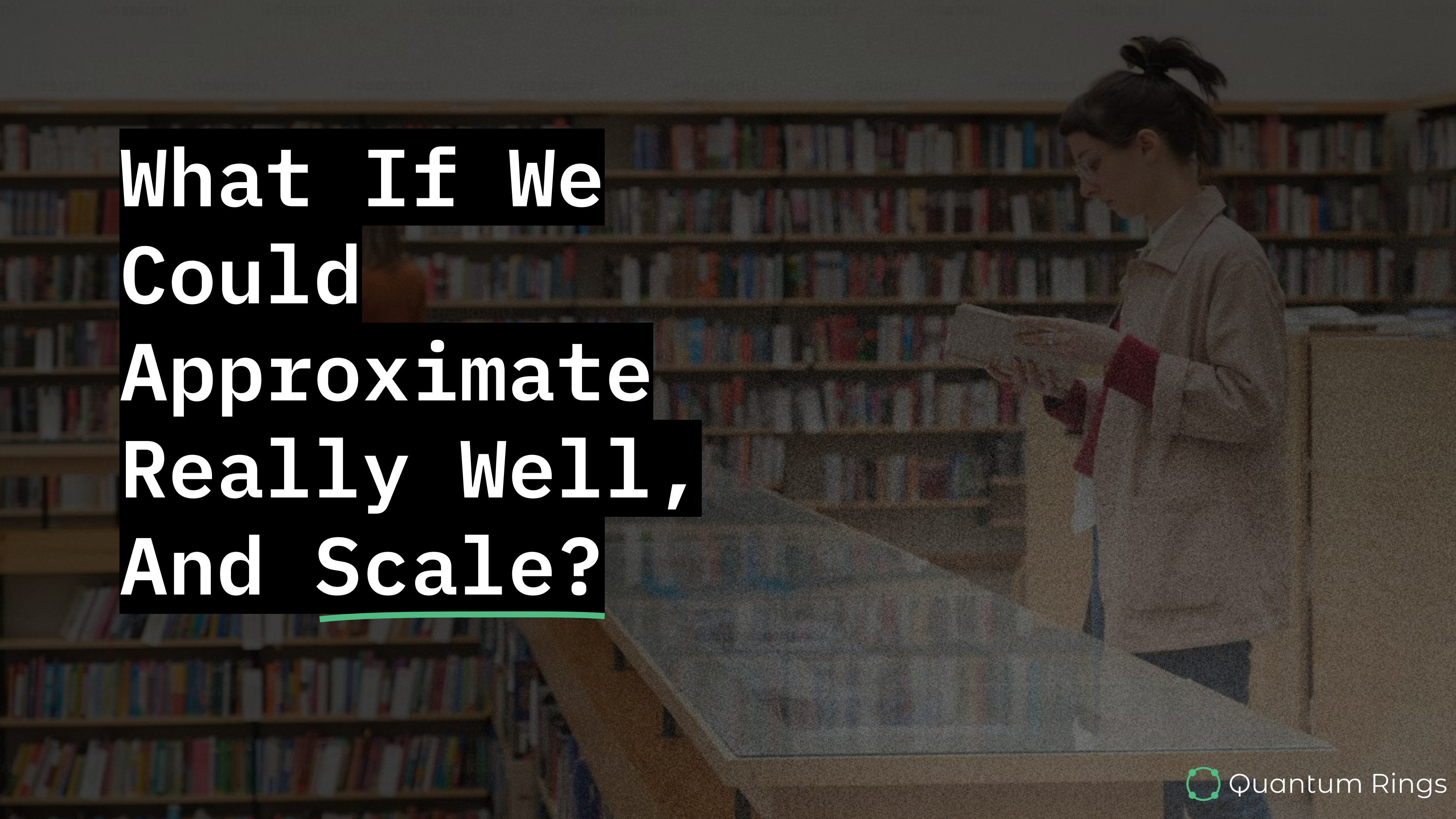
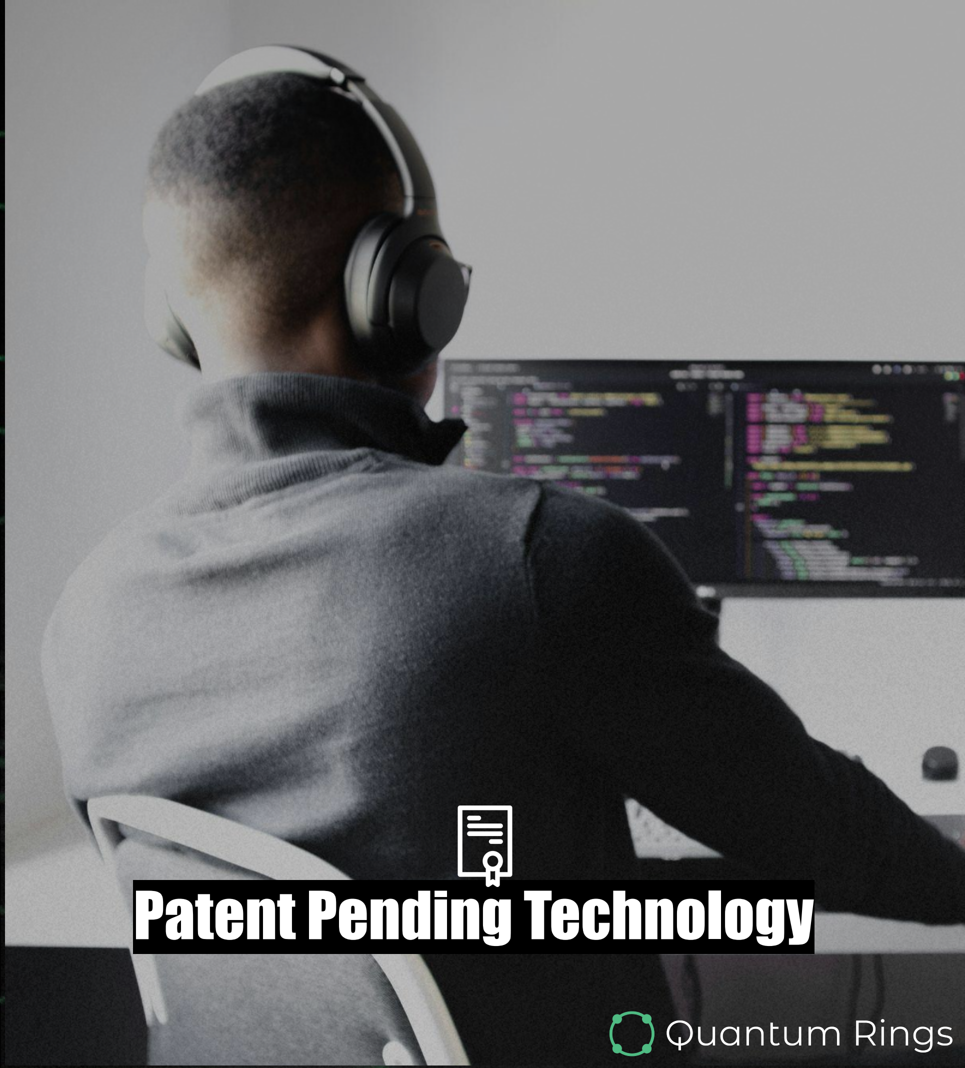
What If We Could Approximate Really Well, And Scale?



Quantum Rings

- √ Hundreds of Qubits
- √ Millions of Ops
- √ High Fidelity
- √ On Your Desktop
- √ In Your Cloud
- √ Today



Quantum Rings

Attracting a Strong Organic Users base

250+ Universities

6,000+ Users

10M+ Circuits Executed

10B+ Gate Operations

Organic Users From:

Massachusetts Institute of Technology
HARVARD UNIVERSITY
Yale
THE UNIVERSITY OF CHICAGO
University of Colorado
ILLINOIS
Arizona State University
THE UNIVERSITY OF ARIZONA
東京大学 THE UNIVERSITY OF TOKYO
qBraid
UNIVERSITY OF CALGARY
POSTECH
KIPU QUANTUM
Artificial Brain
APPLIED QUANTUM SOFTWARE
KPMG
Sandia National Laboratories
EY Building a better working world

And Many More…

Up to ~~128~~ 200
Qubits
Free For
Academic
Use.

What about Quantum Hardware?

Assessment Phase

Algorithm + Problem = Application

QPU Experimentation

1. **Strategy**
   Resource Estimates.
   Time Horizons.

2. **Innovation**
   Experimentation.

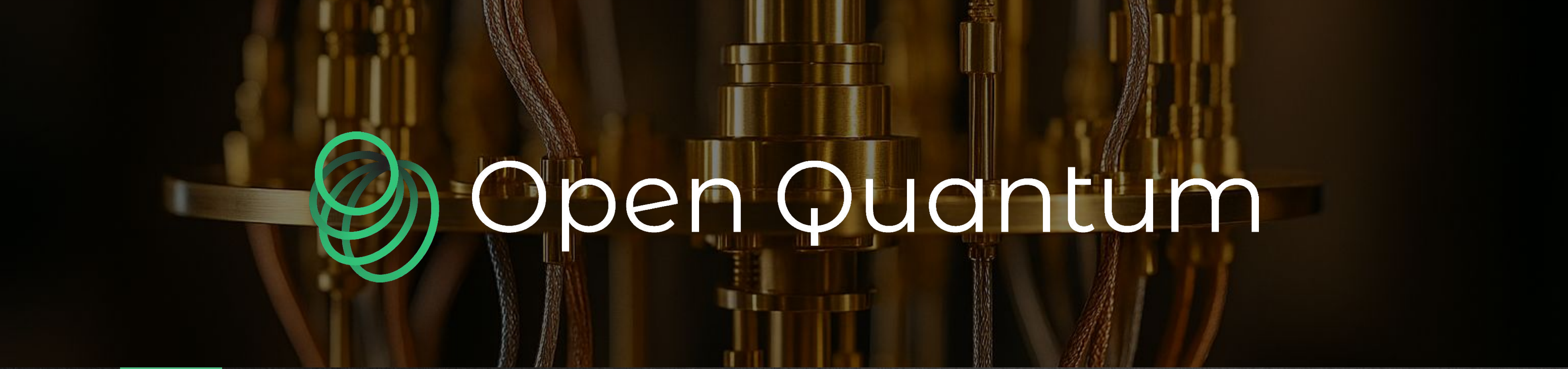
3. **IP Harvesting**

How Might We Make Access to Quantum Compute Ubiquitous?

Quantum Rings

Open Quantum

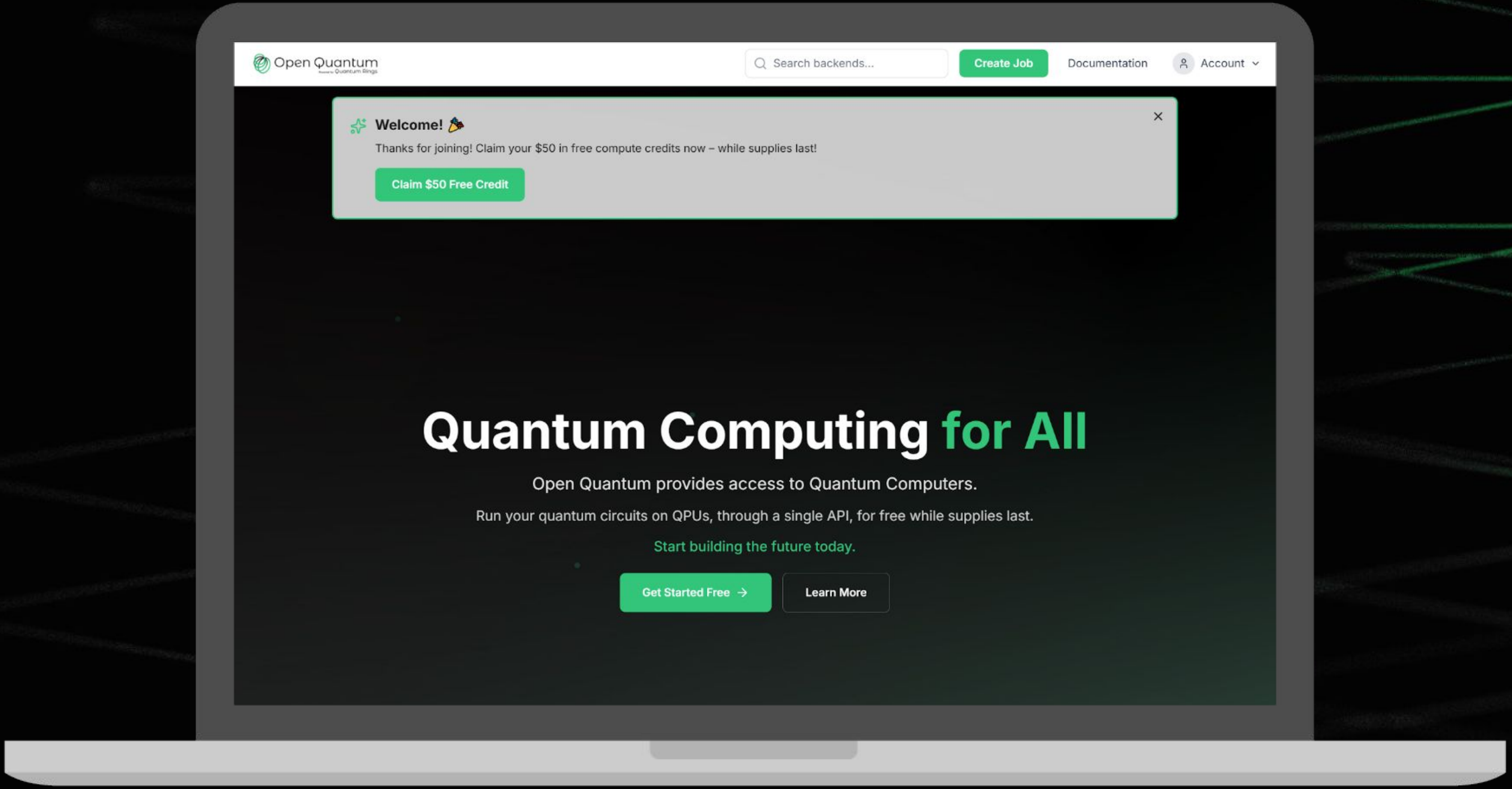
rigetti IONQ IQM aws qBraid



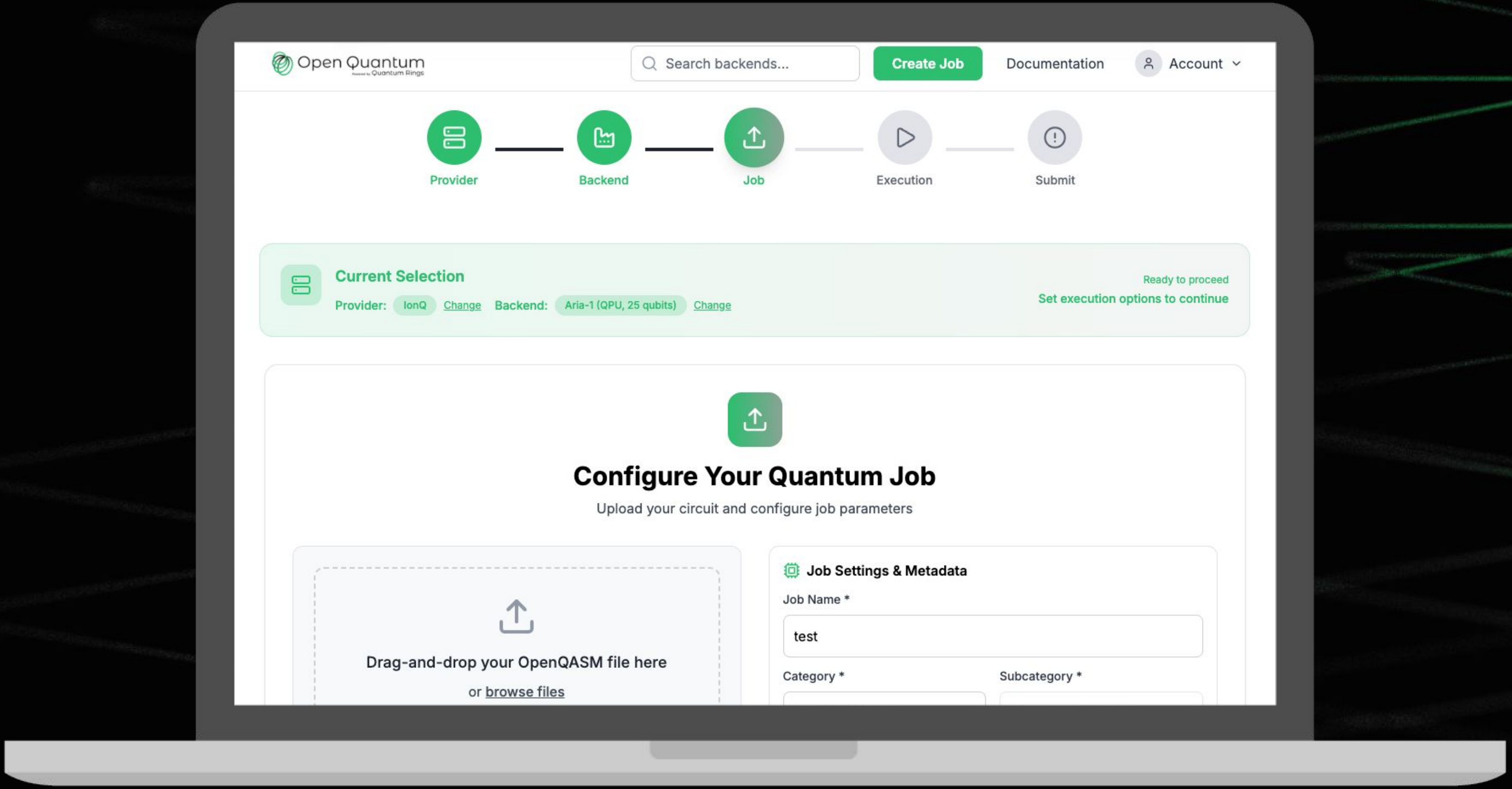
~~$5K-$15k/hr~~ → **Free**

~~12-24hr wait~~ → **0-15 min**

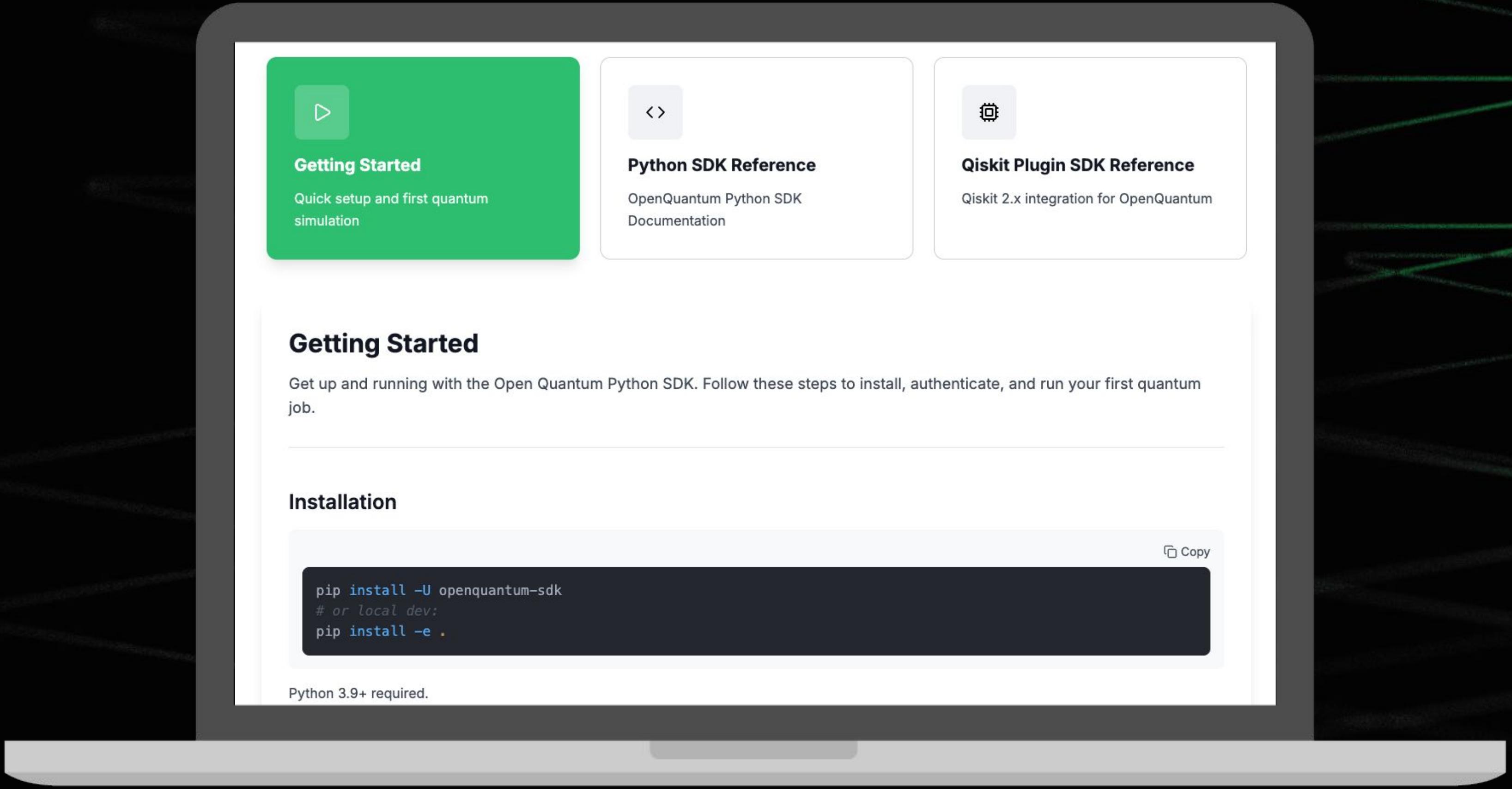
Free Quantum Compute



Quantum Rings



Quantum Rings



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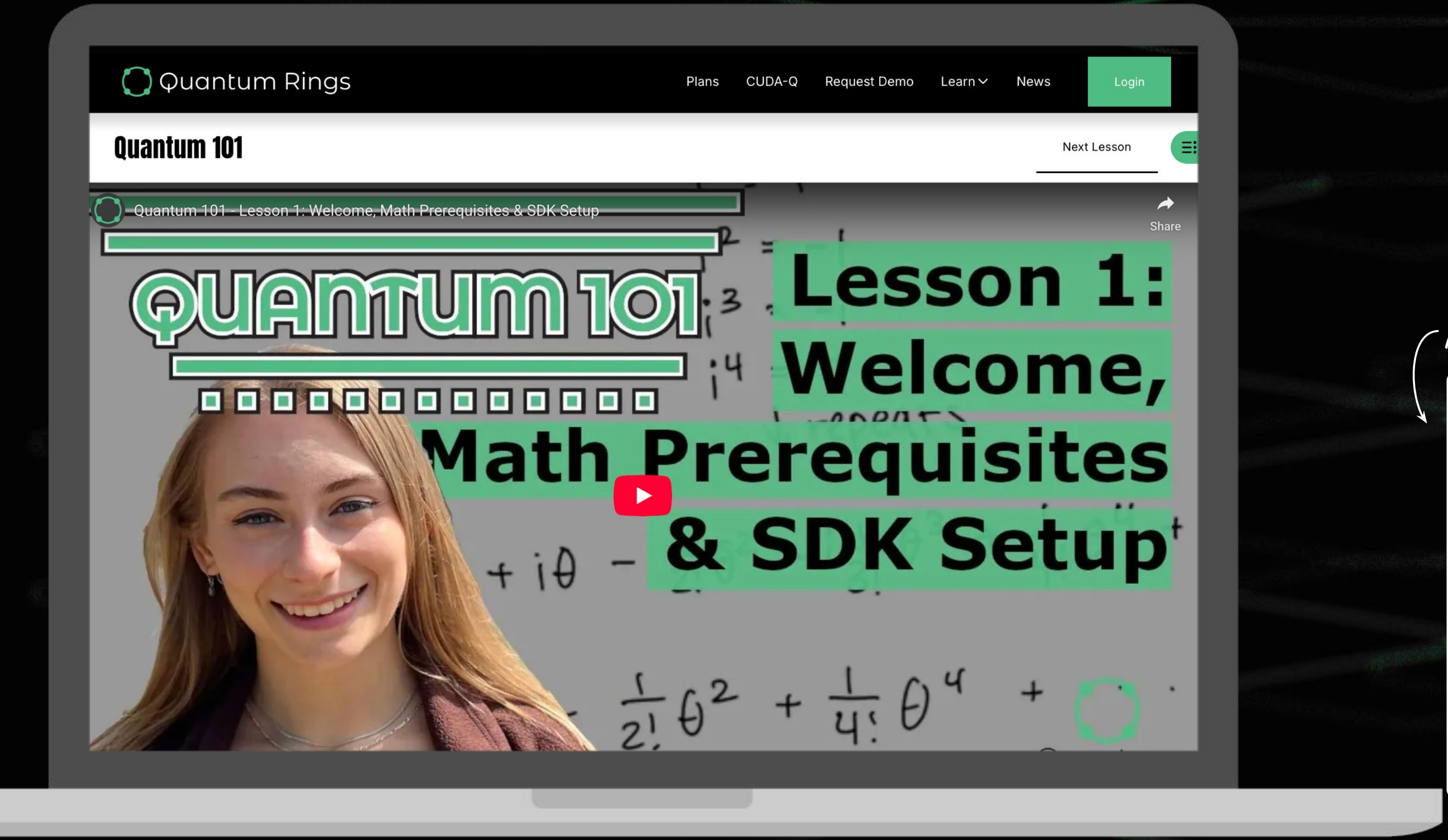
Looking to Get Started Coding Quantum?



Free 14 Week Course

Quantum Rings

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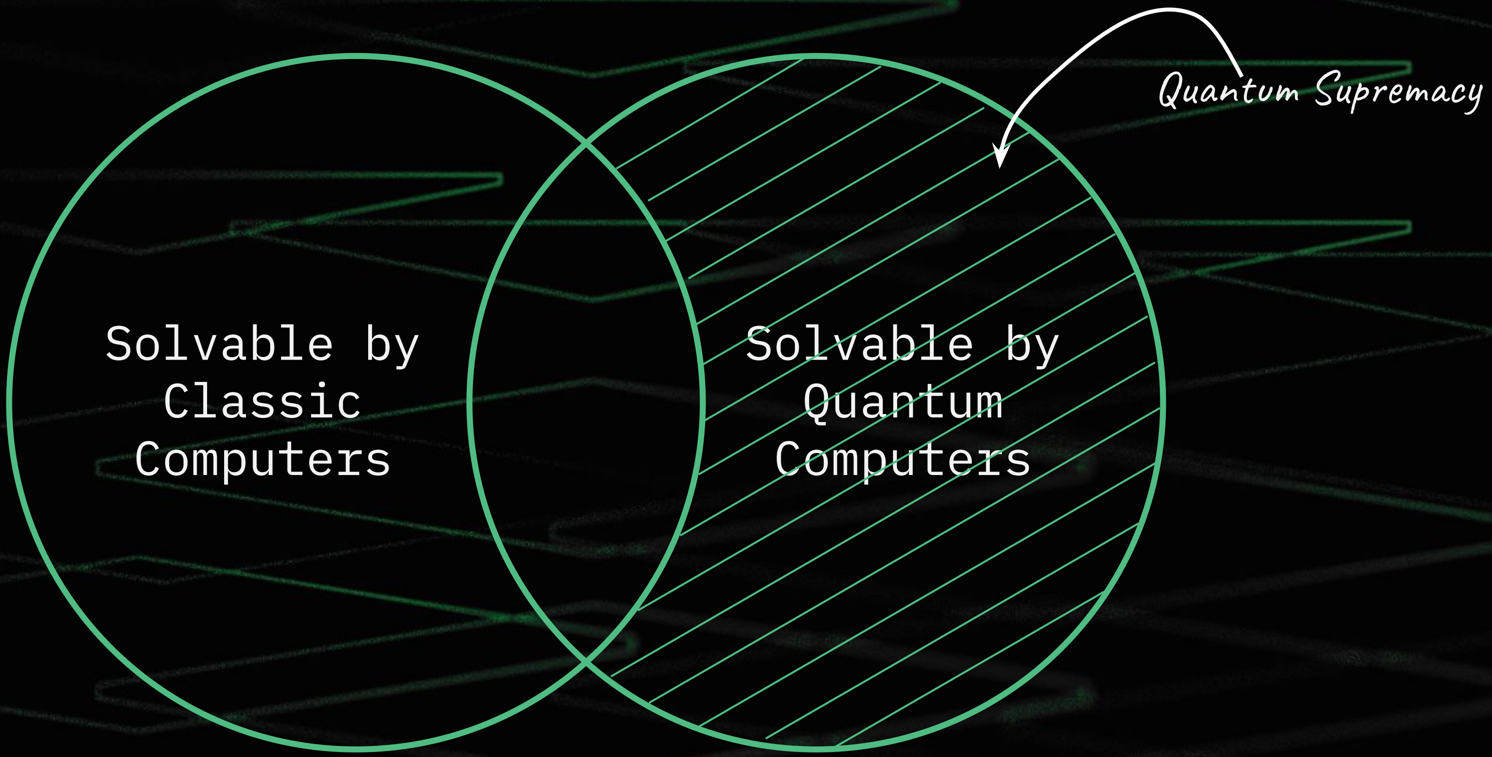
Simulation at Scale

Quantum Rings


Solvable by Classic Computers
Solvable by Quantum Computers
Quantum Supremacy

In 2019 Google Claimed to Have Achieved it!

Quantum supremacy using a

nature.com/articles/s41586-019-1666-5

nature

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Article | Published: 23 October 2019

Quantum supremacy using a programmable superconducting processor

Frank Arute, Kunal Arya, Ryan Babbush, Dave Bacon, Joseph C. Bardin, Rami Barends, Rupak Biswas, Sergio Boixo, Fernando G. S. L. Brandao, David A. Buell, Brian Burkett, Yu Chen, Zijun Chen, Ben Chiaro, Roberto Collins, William Courtney, Andrew Dunsworth, Edward Farhi, Brooks Foxen, Austin Fowler, Craig Gidney, Marissa Giustina, Rob Graff, Keith Guerin, … John M. Martinis + Show authors

Nature **574**, 505–510 (2019) | Cite this article

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Associated content

Quantum computing takes flight

William D. Oliver

Nature **News & Views** 23 Oct 2019

Quantum supremacy: A three minute guide

Elizabeth Gibney

Nature **Nature Video** 05 Nov 2019

Sections | Figures | References

Abstract

The promise of quantum computers is that certain computational tasks might be executed exponentially faster on a quantum processor than on a classical processor[1]. A fundamental challenge is to build a high-fidelity processor capable of running quantum algorithms in an exponentially large computational space. Here we report the use of a processor with programmable superconducting qubits[2,3,4,5,6,7] to create quantum states on 53 qubits, corresponding to a computational state-space of dimension 2^{53} (about 10^{16}). Measurements from repeated experiments sample the resulting probability distribution, which we verify

We Simulated the Experiment and Published The Result

[2411.12131] Empowering Lar

arxiv.org/abs/2411.12131

Cornell University

We gratefully acknowledge support from the Simons Foundation, member institutions, and all contributors. Donate

arXiv > quant-ph > arXiv:2411.12131

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Quantum Physics

[Submitted on 19 Nov 2024]

Empowering Large Scale Quantum Circuit Development: Effective Simulation of Sycamore Circuits

Venkateswaran Kasirajan, Torey Battelle, Bob Wold

Simulating quantum systems using classical computing equipment has been a significant research focus. This work demonstrates that circuits as large and complex as the random circuit sampling (RCS) circuits published as a part of Google's pioneering work [4-7] claiming quantum supremacy can be effectively simulated with high fidelity on classical systems commonly available to developers, using the universal quantum simulator included in the Quantum Rings SDK, making this advancement accessible to everyone. This study achieved an average linear cross-entropy benchmarking (XEB) score of 0.678, indicating a strong correlation with ideal quantum simulation and exceeding the XEB values currently reported for the same circuits today while completing circuit execution in a reasonable timeframe. This capability empowers researchers and developers to build, debug, and execute large-scale quantum circuits ahead of the general availability of low-error rate quantum computers and invent new quantum algorithms or deploy commercial-grade applications.

Comments: 10 pages, 5 figures
Subjects: **Quantum Physics (quant-ph)**; Computational Complexity (cs.CC); Emerging Technologies (cs.ET)
ACM classes: B.8.1; B.8.2; C.4; F.2.1; G.3; I.6.0
Cite as: arXiv:2411.12131 **[quant-ph]**
(or arXiv:2411.12131v1 **[quant-ph]** for this version)
https://doi.org/10.48550/arXiv.2411.12131

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From: Venkateswaran Kasirajan [view email]
[v1] Tue, 19 Nov 2024 00:13:44 UTC (712 KB)

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Arizona State University

NCSA

Quantum Rings

Significantly More Accurate than QPU

Rings

Calculate Quantum State

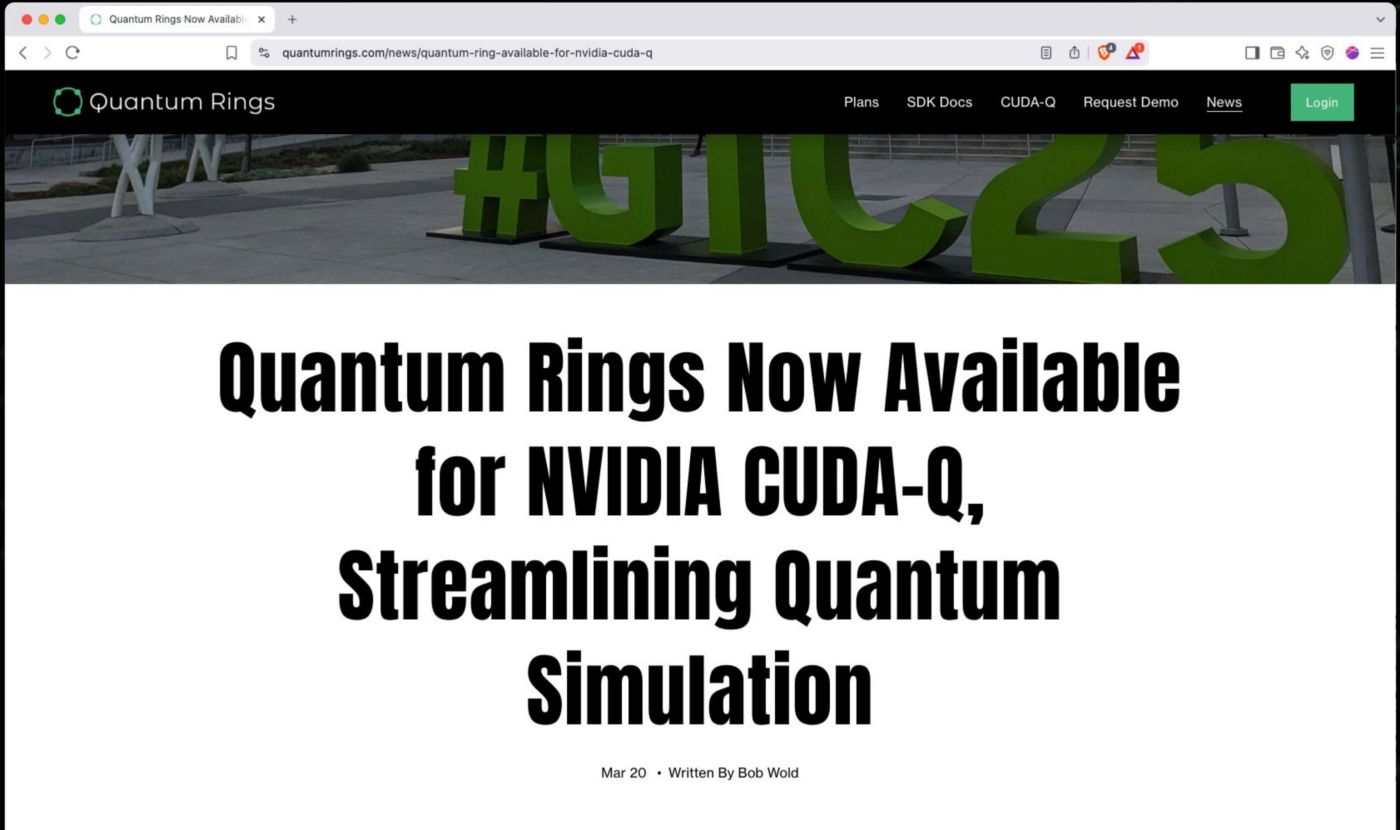
2.5 M SHOTS

2.5 Days on CPU w/32 GB RAM

Full Paper:
https://arxiv.org/abs/2411.12131

GitHub:
https://github.com/Quantum-Rings/quantum-rings-rcs-benchmarking

We Partnered with NVIDIA and Added GPU Support



Quantum Rings Now Available for NVIDIA CUDA-Q, Streamlining Quantum Simulation

Mar 20 • Written By Bob Wold

Quantum Rings

Calculate Quantum State

2.5 M SHOTS

2.5 Days on CPU w/32 GB RAM

~12:30 on GPU

Calculate Quantum State

25K SHOTS
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25K SHOTS

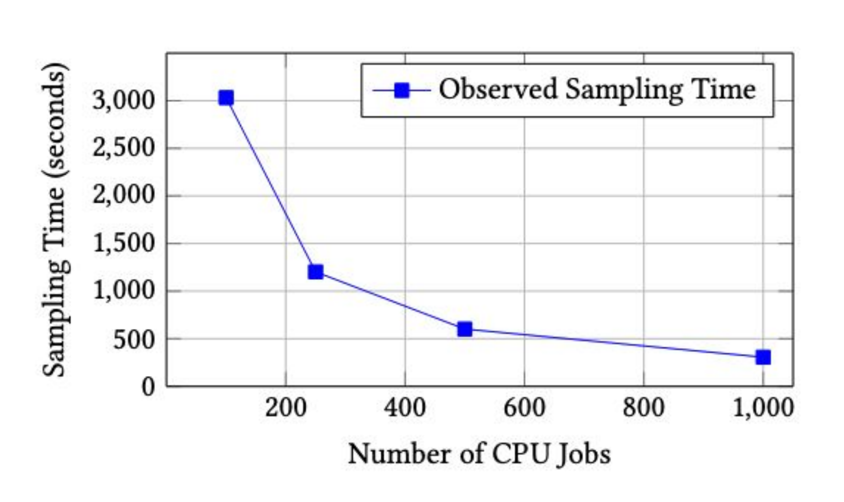
100x CPU instances
16GB RAM
8 CPU cores
...~50 min each

Total Experiment: 01:15:36

6.95×10^7 Speedup v. 10,000 yrs

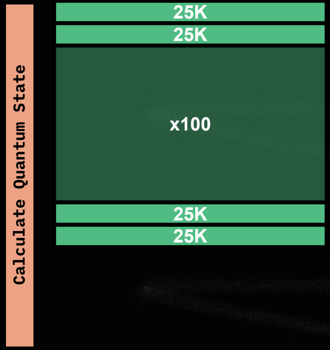
...and only 22x slower than the QPU

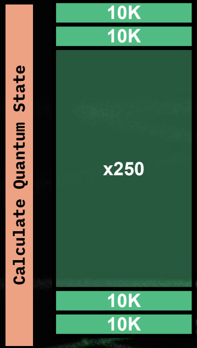
Time is Reduced With Additional CPU Nodes



Jobs	Shots Per Job	Sampling Time (sec)
100	**25,000**	**3,032.0165**
250	10,000	1,202.5353
500	5,000	601.8025
1000	2,500	306.3081

Quantum Rings





Jobs	Shots Per Job	Sampling Time (sec)
100	25,000	3,032.0165
250	10,000	1,202.5353
500	5,000	601.8025
1000	2,500	306.3081

Total Experiment time: 00:17:36

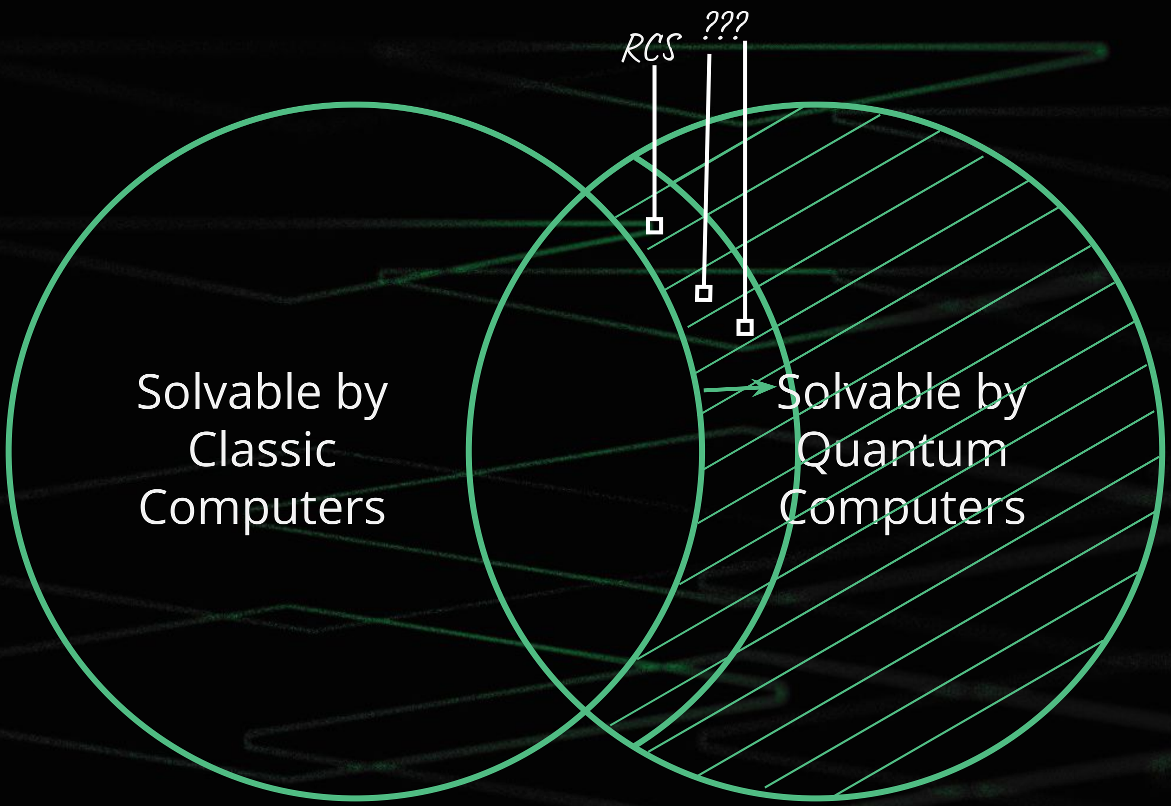
Only 5x slower than the Quantum Computer!

Nothing about this simulation is specific to RCS

RCS
Peaked Circuits
VQE

Shor's
HHL
QML

…and many more


RCS
???
Solvable by Classic Computers
Solvable by Quantum Computers

Beyond RCS: 127 Qubit QAOA

IBM

127 Qubit Optimization[1]

8 minutes on Real Quantum Computer = $768 in cost

30 Minutes on a Laptop

More accurate results
No incremental costs

```
#QiskitRuntimeService.save_account(channel="ibm_quantum", token="<MY_IBM_QUANTUM_TOKEN>", overwrite=True, set_as_
#service = QiskitRuntimeService(channel='ibm_quantum')
provider = QuantumRingsProvider(
    token='rings-1[redacted]AWJn',
    name='bob@quantumrings.com'
)
provider.save_account(provider)

#backend = service.least_busy(min_num_qubits=127)
backend = QrBackendV2(provider=provider, num_qubits = circuit.num_qubits)
```

```
#from qiskit_ibm_runtime import Session, EstimatorV2 as Estimator
from quantumrings.toolkit.qiskit import QrEstimatorV2 as Estimator

#estimator = Estimator(mode=session)
estimator = Estimator(backend=backend)
```

```
[ ]: #from qiskit_ibm_runtime import SamplerV2 as Sampler
     from quantumrings.toolkit.qiskit import QrSamplerV2 as Sampler

     #sampler = Sampler(mode=backend)
     sampler = Sampler(backend=backend)
```

[1] Solve utility-scale quantum optimization problems, IBM

Quantum Rings

Beyond RCS: Shor's Algorithm

39

Largest Bit Integer Factored using Shor's Algorithm published so far[1]

40

45

48

For Perspective:

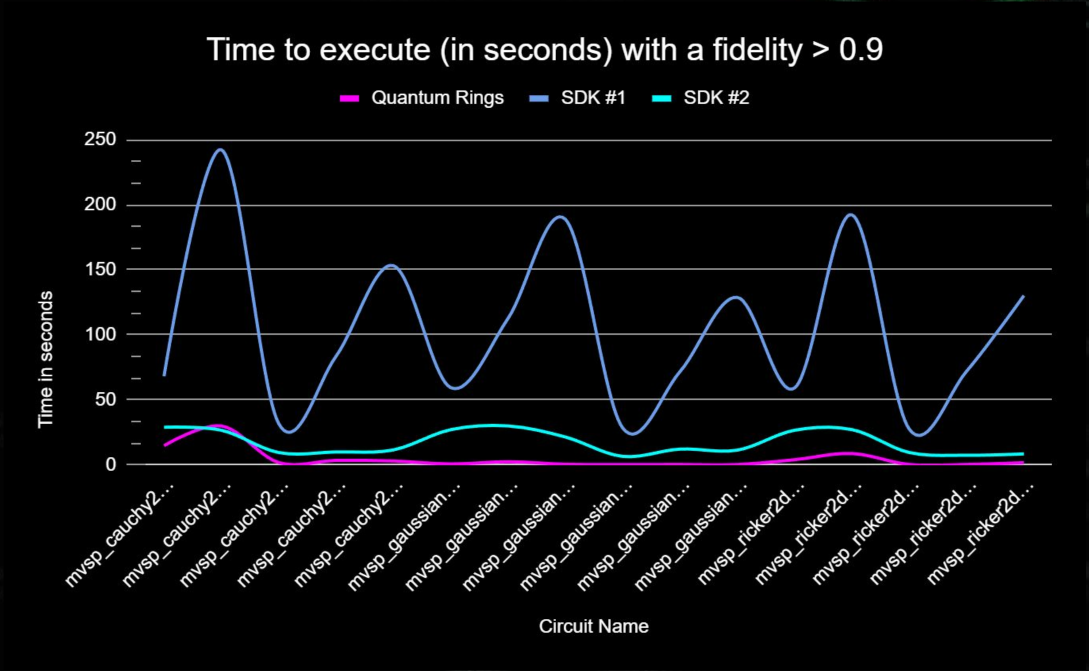
549, 755,813,701 (39 bit)
using 2048 GPUs of JUWELS Booster (Willsch et al. 2024)[1]

255,975,740,711,783 (48 bits)
using a single, CPU-Only Laptop running Quantum Rings

[1] https://arxiv.org/abs/2410.14397, (Willsch et al. 2024)

Quantum Rings

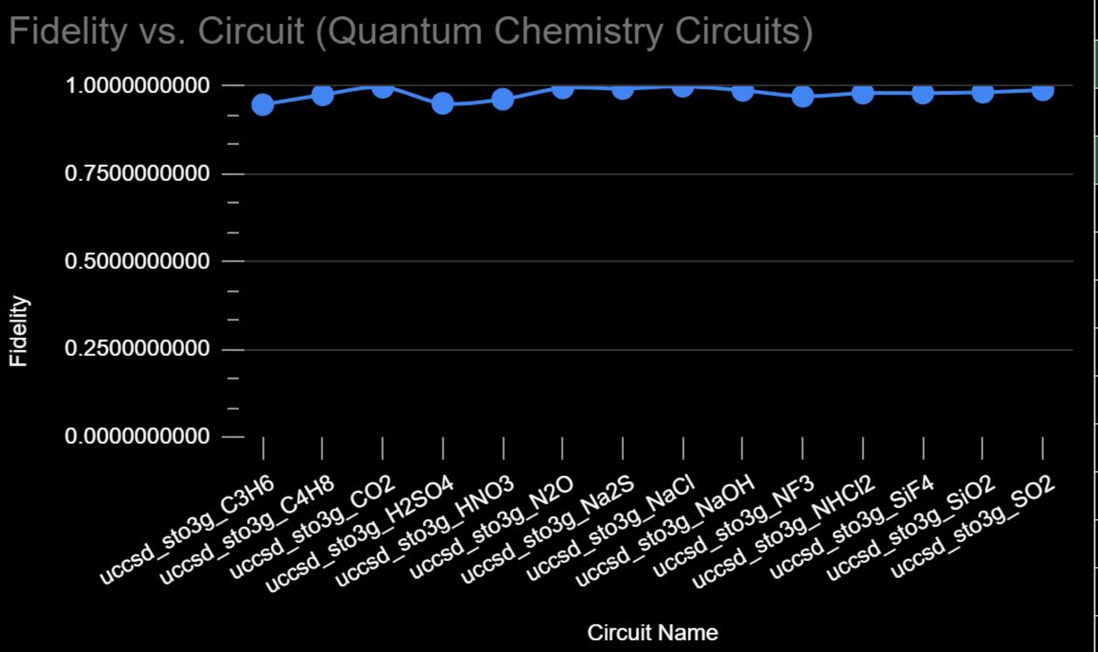
How Fast Is It?



Circuit	Gates	Qubits
mvsp_cauchy2d_fourier_d10_n10	10,770	30
mvsp_cauchy2d_fourier_d10_n40	12,168	90
mvsp_cauchy2d_fourier_d5_n10	2,991	28
mvsp_cauchy2d_fourier_d5_n25	3,651	58
mvsp_cauchy2d_fourier_d5_n40	4,131	88
mvsp_gaussian2d_fourier_d10_n10	10,771	30
mvsp_gaussian2d_fourier_d10_n25	11,581	60
mvsp_gaussian2d_fourier_d10_n40	12,169	90
mvsp_gaussian2d_fourier_d5_n10	2,991	28
mvsp_gaussian2d_fourier_d5_n25	3,651	58
mvsp_gaussian2d_fourier_d5_n40	4,131	88
mvsp_ricker2d_fourier_d10_n10	10,771	30
mvsp_ricker2d_fourier_d10_n40	12,169	90
mvsp_ricker2d_fourier_d5_n10	2,991	28
mvsp_ricker2d_fourier_d5_n25	3,651	58
mvsp_ricker2d_fourier_d5_n40	4,131	88

(*- Measured in CPU ONLY mode on an NVIDIA GB10 (GPU acceleration not used), data to be published shortly)

How Accurate Is It?



Circuit	Gate Count	Qubits	Fidelity
uccsd_sto3g_C3H6	772,149	42	0.9470610000
uccsd_sto3g_C4H8	2,822,536	56	0.9750780000
uccsd_sto3g_CO2	64,734	30	0.9965780000
uccsd_sto3g_H2SO4	2,826,152	62	0.9508490000
uccsd_sto3g_HNO3	862,165	42	0.9626930000
uccsd_sto3g_N2O	127,543	30	0.9946820000
uccsd_sto3g_Na2S	1,195,416	54	0.9927200000
uccsd_sto3g_NaCl	133,899	36	0.9994330000
uccsd_sto3g_NaOH	167,909	30	0.9878000000
uccsd_sto3g_NF3	277,073	40	0.9710720000
uccsd_sto3g_NHCl2	346,374	48	0.9799630000
uccsd_sto3g_SiF4	618,680	58	0.9800580000
uccsd_sto3g_SiO2	119,403	38	0.9824870000
uccsd_sto3g_SO2	119,636	38	0.9891400000

(*- Simulated on a single H100 class NVIDIA GPU, data from a previous result
https://github.com/CQCL/public_tn_sim_challenge_results/tree/main/QuantumRings)

“How Big”?
“How Fast”?
“How Accurate”?
“How Much Memory”?

…”It Depends”

|010⟩

The Challenge

Quantum Rings

Why circuit structure matters

- Tensor-network simulators are fast... *until they aren’t*
- Entanglement growth & complex rotations drive memory and runtime
- Simulators make approximations to speed the calculation while still providing quality results
- There’s no single “best” setting — it depends on the circuit

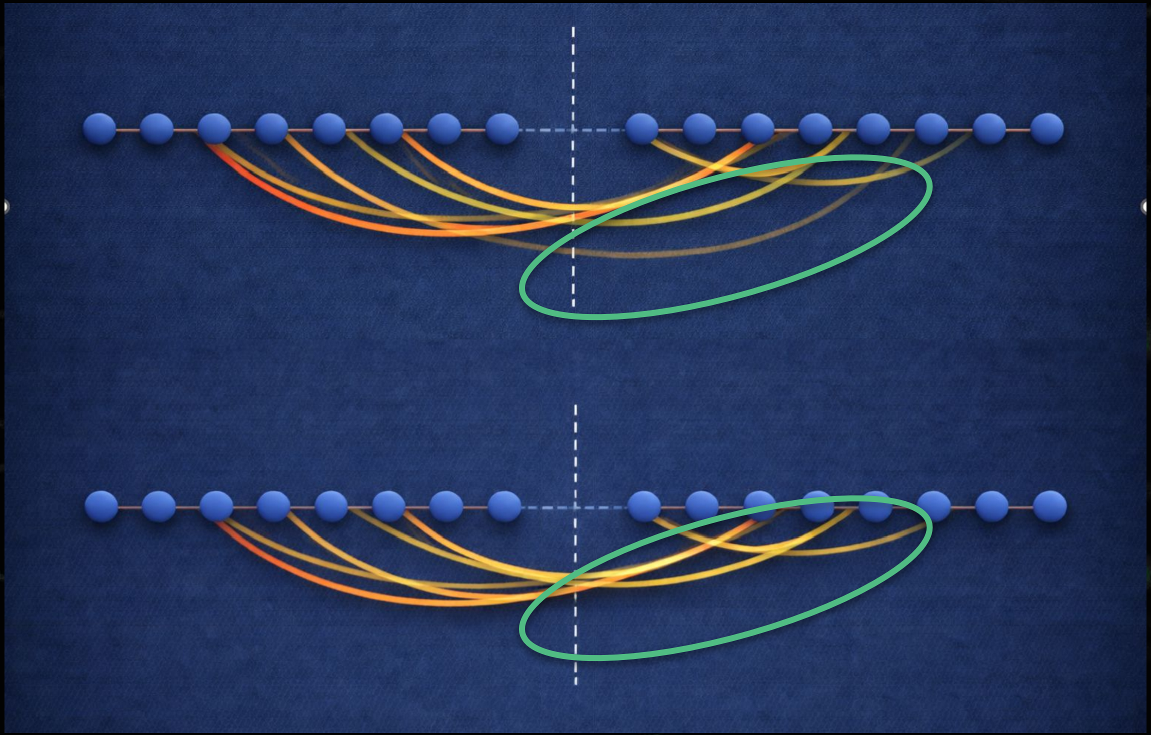
The Challenge

- Analyze the circuit structure to understand how it will impact the calculation
- Tune the available knobs
  - CPU vs GPU
  - Single vs Double precision
  - Threshold
- Create a model that will predict simulation performance
  - Runtime
  - Fidelity

The Knobs

- GPU vs CPU
  - GPU processing can be faster than CPU. But it comes with overhead so the walltime can be slower on small problems. Circuit construction vs execution time will also vary with processor choice.
  - When should you use a CPU or GPU?
- Single vs Double
  - Single precision calculations are faster than double but have less precision and can reduce fidelity.
  - When is it worth using double precision?
- Threshold
  - Limits the size of the calculation in the simulation
  - Too low costs fidelity. Too high costs time.

What does Threshold do?



Threshold = Compression

Quantum Rings

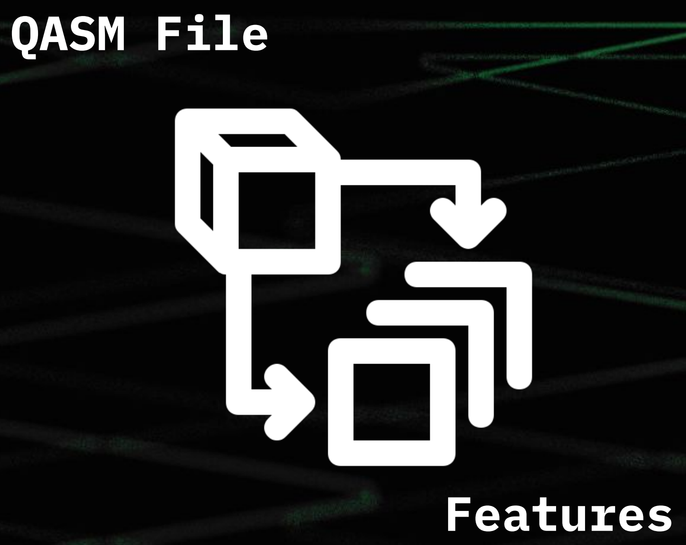
What we give you

- A curated collection of **OpenQASM 2 circuits** giving a range of sizes and entanglement structures
- A **labeled training dataset** derived from real simulator runs
- An **unlabeled holdout task** list specifying required prediction targets (circuits withheld)
- **Documentation** describing the simulator model, threshold concept, and scoring rules
- A **Google Forms** submission interface and local format validator

The Training Data

- 36 distinct circuits
- For each circuit multiple executions
  - CPU & GPU
  - Single & Double precision
  - Threshold [1, 2, 4, 8, …, 256]
- For each execution we provide:
  - Runtime
    - Single shot fidelity
    - 10,000 shot forward runs
  - Fidelity
    - Mirror fidelity – $U \cdot U^\dagger$

Feature Engineering is Key



Feature Engineering Ideas

- Gate Counts
- 2Q Gate Density
- Circuit Depth
- Interaction Graph Width
- Entanglement Depth Proxies
- Long-range Gates
- Cross-cut Pressure
- …and Many More!

A Little Research, Could Go A Long Way!

Quantum Rings

What you submit

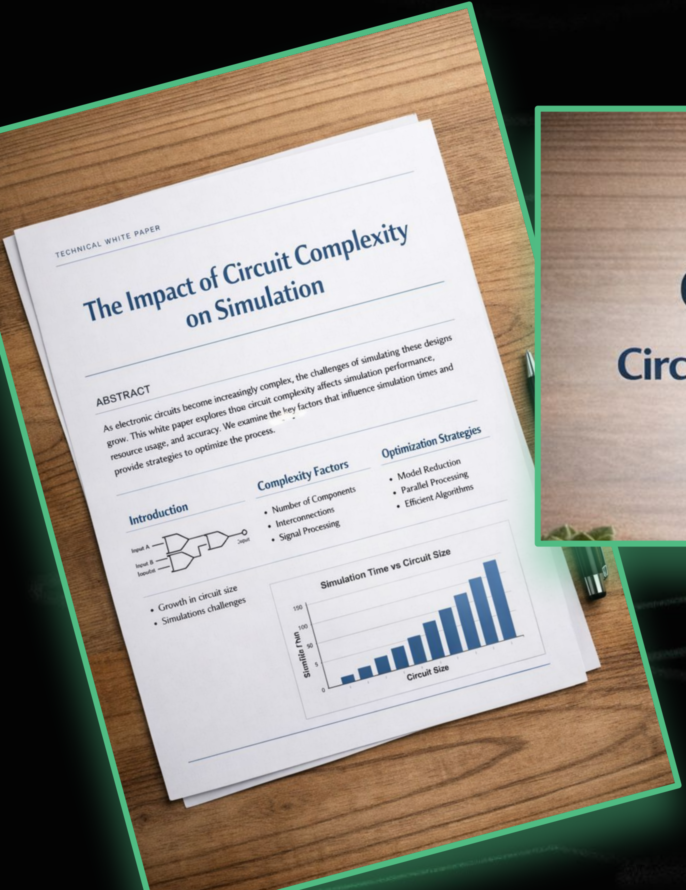
- Model, `predict.py`, and supporting code (zip)
  - **predicted_threshold_min** – one of the ladder rungs (1, 2, 4, 8, 16, 32, 64, 128, 256)
  - **predicted_forward_wall_s** – predicted wall-clock seconds for the **10,000-shot forward run** at that rung
- Write Up
  - Describing your model
- Presentation
  - Including a live run of your model to on the holdout circuits

How you’re scored

- Combined Score
  - Presentation
  - Model Predictions
  - 5 Holdout Circuits
    - 20 tasks (5 circuits, CPU/GPU, Single/Double)
  - Runtime and Fidelity
- **Predictions must achieve the mirror fidelity target** >= 0.99
  - missing the fidelity results in a zero score for the task
- Timing
  - normalized and scored symmetrically with equal weight to over/under

Quantum Rings

Too Easy? Take if Further!



Prizes & Recognition

$200 Quantum Compute Credits

(Max $1,000 for team)

Plus:

- **Blog Post**, Highlighting the winning team, and mentioning the winners by name
- **LinkedIn Post** from Quantum Rings, tagging the winners
- **Guaranteed Interview** for Quantum Internships, Summer 2026

NVIDIA prizes for the MIT IQuHack

Four ways to win!

NVIDIA Challenge Awards

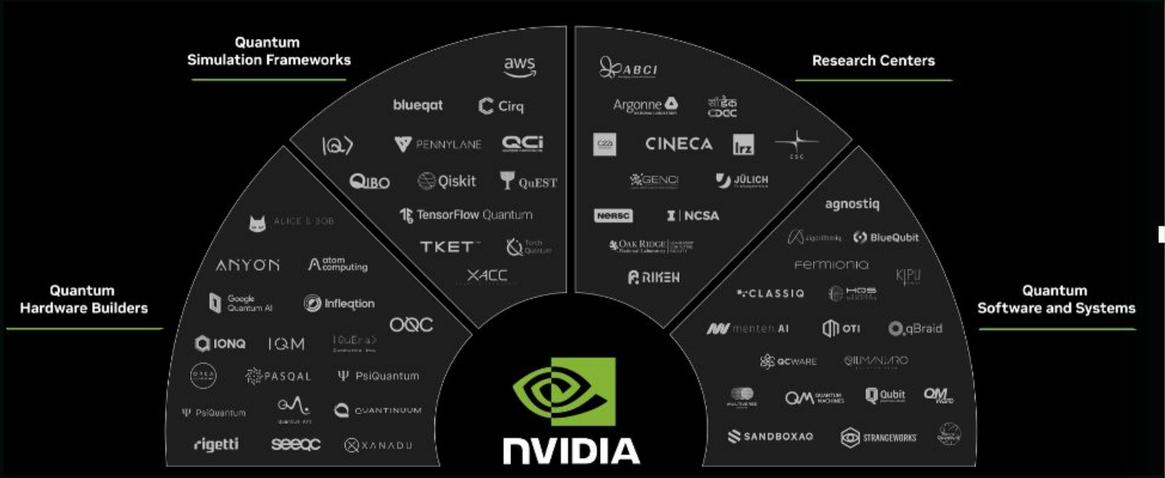
Eligibility: complete the NVIDIA challenge

- 1st Prize 400 brev credits /person*
- 2nd Prize 300 brev credits/person*
- 3rd Prize 100/ brev credits person*

NVIDIA Ecosystem Award

Eligibility: Use CUDA-Q or NVIDIA tech in one of the other challenges (e.g. IQM, Quantum Rings). Submit your code and a short description of your project and use of NVIDIA tech to the NVIDIA discord channel

- One prize: 200 brev credits/person*
- Other challenges where CUDA-Q or NVIDIA tech may be useful: IQM, Quantum Rings



* Max 5 people per team

NVIDIA

Quantum Rings

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Thanks

+

Q&A

Free Quantum Compute

Free 14-Week Course

Free Large Scale Simulation

Promocode

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Summer Internships